Government College of Engineering, Chhatrapati Sambhajinagar

B. Tech. First Year In Semester Examination 1

ETIKS1001: INDIAN KNOWLEDGE SYSTEM (CSE BRANCH)

Date: October 25, 2024 **Time: 45 MINUTES** **Max. Marks: 10**

Q. N.	Solve any three.	CO	B.T. Level	Marks
1	**A.** Early chemical practices and transformations are included in _____. a. Alchemy **b**. Kavya **c**. Psychology **d**. Dharma **B.** Key schools of Indian Philosophy include _____. a. Vedanta **b**.Samkhya **c**. Yoga **d**. All of these	1,2	K1	02
2	**A.** Define Veda. Explain all Vedas in detail. OR **B.** What are potential fallacies in the reasoning process.	1,2	K2	04
3	**A.** Define knowledge. Describe knowledge triangle in IKS. OR **B.** Draw and explain classification framework of Indian Knowledge System	2 1	K1	04

Note: CO: Course Outcome; B.T. Level: Bloom's Taxonomy Level

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Government College of Engineering Aurangabad
Chhatrapati Sambhajinagar
(An Autonomous Institute of Government of Maharashtra)
FY B.Tech CSE Examination

Class Test – II
***CSIKS 1003*: Indian Knowledge System**
Time: One Hour Date of Examination: 26-12-2024 *Max. Marks: 10*

"Verify the course code and check whether you have got the correct question paper"
N.B:-Assume suitable data if necessary and state it clearly
2. *Use of non-programmable calculator is allowed 3.Attempt all questions in sequence*

		Course Outcomes (COs)	Bloom's Taxonomy Levels(BTLs)	Marks
Q1.	**Multiple choice questions**			
	1) _____ made contributions to the understanding of numbers and arithmetic operations. a. Aryabhatta **b**. Brahmagupta **c**. Bhaskara II **d**. All of these	CO1, CO2	K1	2
	2.The _____ Indian text significantly describes methods for calculating eclipses and the motion of planets. a. Vedas **b**. Upanishads **c**.Siddantas **d**. Chandah-sastra			
Q2.	**Attempt any one**	CO1, CO2	K2	4
	1)Give importance of decimal representation.			
	2)Explain salient features of the Indian numeral system.			
Q3.	**Attempt any one**			
	1)Describe significant contributions of following mathematicians: a.Aryabhatta b.Bhaskara II 2)Describe contributions to Binary Mathematics and Magic Squares in India.	CO2, CO3	K1	4

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Government College of Engineering, Chhatrapati Sambhajinagar

(An Autonomous Institute of Government of Maharashtra)

End Semester Examination November-December 2024

First Year B.Tech CSE

Time: 1 Hour 30 Mins **CSIKS1003: Indian Knowledge System** ***Max. Marks: 30***

16 JAN 2025

"Verify the Course Code and check whether you have got the Correct Question Paper"

1. ***Attempt all questions.***
2. ***Figures to the right indicate full marks.***

Q1	**Attempt all**	*CO*	*B.T. Level*	*Marks*
A.	What are the elements of Indian calendar? Write note on Equinotical points and precession of the Equinoxes.	*4*	*K1*	*06*
B.	Write a short note on Sanskrit and give its significance.	*2*	*K2*	*06*
Q2	**Attempt any TWO**	*CO*	*B.T. Level*	*Marks*
A.	Explain Phonetics in Sanskrit.	*2*	*K2*	*06*
B.	What are the unique aspects of Indian Mathematics.	*3*	*K2*	*06*
C.	Give importance of (i) Decimal Representation (ii) Discovery of Zero	*3*	*K1*	*06*
Q3	**Attempt any ONE**	*CO*	*B.T. Level*	*Marks*
A.	Explain importance of IKS in detail.	*1*	*K2*	*06*
B.	Write note on (i)Atharvaveda (ii)Yajurveda	*1*	*K1*	*06*

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Government College of Engineering, Aurangabad
Chatrapati Sambhajinagar
(An Autonomous Institute of Government of Maharashtra)
B.Tech F.Y. Civil & CSE (NEP)
Class Test – I
CEESC1001: Basics of Civil Engineering
Date of Examination: 24nd Oct 2024

Time: 45 minutes *Max. Marks: 10*

"Verify the course code and check whether you have got the correct question paper"

Q. 1	**Attempt any Two.**	CO	BTL	Marks
1.	What is Principle of Planning. Explain any three.	CO1	K1	3
2.	Explain following technical terms- a. Carpet area b. Floor space Index (FSI) c. Build-up area	C01	K1	3
3.	Explain in details any three types of loads	C01	K2	3
Q.2	**Attempt Any One.**			
1	Explain Load bearing structure and Frame structure with neat sketch.	CO1	K1	4
2.	Define Important terms of Masonry with diagram	CO1	K2	4

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Government College of Engineering, Aurangabad
Chatrapati Sambhajinagar
(An Autonomous Institute of Government of Maharashtra)
B. Tech F.Y. Civil & CSE (NEP)
Class Test – II
CEESC1001: Basics of Civil Engineering
Date of Examination: 24nd Dec 2024

Time: 45 minutes *Max. Marks: 10*

"Verify the course code and check whether you have got the correct question paper"

Q.1	**Attempt any Two.**	**CO**	**BTL**	**Marks**
1.	Distinguish between Rigid pavement and Flexible pavement.	CO3	K2	3
2.	Explain Remote sensing and GIS (Geographical information System).	C02	K1	3
3.	Explain in detail Brick & also write classification & field tests of bricks in detail.	C03	K2	3
Q.2	**Attempt Any Two.**			
1	Convert the following WCB to QB/RB b. 53° 55' b. 175° 45' d. 222° 35' d. 305° 55'	CO2	K2	2
2.	Convert the following QB/RB To WCB a. N 15° 35' E b. S 36° 37'E c. S 44° 43' W d. N 24° 56' W	CO2	K2	2
3.	Explain Bridge in detail with classification.	CO3	K1	2

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Government College of Engineering, Chhatrapati Sambhajinagar

(An Autonomous Institute of Government of Maharashtra)

F.Y.B.TECH. Civil (NEP) Examination

End Semester Examination Nov-Dec 2023

CEESC 1001 : Basics Of Civil Engineering

13 DEC 2023

Time: 1 Hour 30 mins **Max. Marks: 30**

"Verify the Course Code and check whether you have got the Correct Question Paper"

N.B.

1. ***Attempt all questions.***
2. ***Figures to the right indicate full marks.***
3. ***Assume suitable data if necessary and state it clearly.***
4. ***Use of Non-Programmable Calculator and Data Sheet is allowed.***

Q1	**Attempt any TWO**	***CO***	**B.T. Level**	***Marks***
A	Which points should be considered in the selection of site for: 1. Residential Building 2. Commercial Building	1	2	***05***
B	What are the building Bye- laws? And state the necessities of building Bye laws.	1	4	***05***
C	What is the importance of "Economy" in planning of a building?	1	4	***05***
Q2	**Attempt any TWO**	***CO***	**B.T. Level**	***Marks***
A	Write in detail about remote sensing.	2	2	***05***
B	Describe in brief the temporary adjustments Dumpy level.	2	2	***05***
C	Convert WCB to RB : 1. 136^0 2. $328^0 30'$ 3. $56^0 30'$ 4. $238^0 30'$ 5. 325^0	2	4	***05***
Q3	**Attempt any TWO**	***CO***	**B.T. Level**	***Marks***
A	What are the properties of a good building stone?	3	4	***05***
B	Explain various equipments used for excavation.	4	2	***05***
C	Brief on watershed management.	5	3	***05***

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Government College of Engineering, Aurangabad

(An Autonomous Institute of Government of Maharashtra)

FY. (EC & T and CSE) Examination PPS_CSESC1001: Programming for Problem Solving

Time: ***One Hour*** Date: ***23/10/2024*** Class Test- ***I*** Max. Marks: ***15***

Qu1 Attempt any four from following (two marks to each question) **8 Marks**

a) Compare compiler and interpreter. CO1

b) Define algorithm and flowchart. CO1

c) State the difference between syntax error and logical error.Give suitable example CO1

d) Write the value of d for the following expressions if a=1,b=2,and c=3 CO2

(i) d=a+c*a/b (ii) d=a+c*a%b (iii) d = (a == b) (iv) d = (a != b)

e) Find the error if any in following and rewrite corrected statements CO2

(i) real a=5.5 (ii) char ch="M" (iii) long float =123456 (iv) char name[30]='ABC'

f) Do as directed CO2

(i) Declare variable college to store GECA in it (ii) declare variable gender and assign a value F to it. (iii) Write a c statement to print square of number (iv) write a c statemen to accept values of principal, rate and year from keyboard as 50000, 7.6 and 5

(PTO)

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Qu2 Attempt from following (three marks) **3 Marks**

Give meaning of source file, object file and executable file in 'C' programming. How, when and where these files are created. Explain with suitable example CO1

Or

Write a standard structure of 'C program with suitable example and explain different section and their function. CO1

Qu3 Attempt from following **4 Marks**

Write a program to accepts students name, enrollment number, gender and percentage of marks from keyboard and print the same to give output as following CO2

Name of student : ABC XYZ Enrollment Number : 123456

Gender : M Percent of marks : 78.60

Or

Write a program in 'C' to find the roots of quadratic equation and print the same CO2

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Government College of Engineering, Aurangabad

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FY. (EC & T and CSE)ExaminationPPS_CSESC1001: Programming for ProblemSolving

Time: ***One Hour*** Date: **23/12/2024** Class Test- ***II*** Max. Marks: *15*

Qu1 Attempt any four from following (two marks to each question) **8 Marks**

a) Give syntax of conditional operator. Write code segment to find larger of two given numbers x and y using conditional operator. CO2

b) Give symbol of logical operator. When each shall be used? Write their precedence CO2

c) State the difference between break and continue statement with syntax. execution and example CO3

d) Declare an array appropriately to store the following (i) marks of 75 students (ii) marks of 5 test of 75 students. CO4

e) Define function. State its types with one example of each. Write advantages of function. CO3

f) Write the output of the following CO3

```
int main()
{int a=25;
while(a<=27)
{printf("%d ",a);
    a++;
}
return 0;
}
```

```
int main()
{ inti,j;
for (i=0;i<3;i++)
{ for (j=0;j<3;j++)
printf("%d %d %d", i,j,i*j)
printf("\n");
}
}
```

(PTO)

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Course Outcome (Cos)

Qu2 Attempt from following (threemarks) **3 Marks**

List different storage classes in C. Write i) keyword used(ii) Initial value (iii) scope/ accessibility (iv) Life time/ Longevity (v) Example (vii) application of each type of storage class. CO3

Or

Write a program to print the class of the student based on percentage of marks entered through keyboard. Conditions are (i) Distinction -> percent marks more or equal to 75. (ii) FirstClass -> if marks between sixty and less than 75 (iii) Second Class -> if marks between fifty and less than 60 (iv)Pass Class -> if marks between forty and less than 50 (v) else Fail. Check percent of marks between 0 to 100 else give appropriate message.. CO3

Qu3 Attempt from following **4 Marks**

What is recursion? Write a separate program to find factorial of a number using (i) recursion (ii) iteration. Compare recursion and iteration. CO3

Or

Write a program to accept five numbers from keyboard and find sum, average of these numbers using array. Also print the five numbers as per (i) order of numbers entered through keyboard (ii) reverse order CO4

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Government College of Engineering, Chhatrapati Sambhaji Nagar

(An Autonomous Institute of Government of Maharashtra)

FYBTECH – NEP – SEMESTER – I

End Semester Examination December 2023

CSESC1001:Programing for problem solving

15 DEC 2023

Time: 2 Hours and 30 Minutes ***Max. Marks: 60***

"Verify the Course Code and check whether you have got the Correct Question Paper"

N.B.

1. ***Attempt all questions.***
2. ***Figures to the right indicate full marks.***
3. ***Assume suitable data if necessary and state it clearly.***
4. ***Use of Non-Programmable Calculator and Data Sheet is allowed.***

Q1	**Attempt any two**	***CO***	***B.T. Level***	***Marks***
A	Describe architecture of a computer system in detail	*1*	*1*	*06*
B	Define algorithms and flowcharts in detail. Write an algorithm and flowchart for adding two numbers.	*2*	*2*	*06*
C	Explain standard Input output in C.	*2*	*1,2*	*06*
Q2	**Attempt any two**	***CO***	***B.T. Level***	***Marks***
A	Summarize the different data types in C with examples.	*2*	*2*	*06*
B	Enlist and explain the following operators with example: 1. Arithmetic operators 2. Logical operators 3. Relational operators	*2*	*2*	*06*
C	Write a program to accept a character and convert into an integer and display the value.	*3*	*2*	*06*
Q3	**Attempt any two**	***CO***	***B.T. Level***	***Marks***
A	Describe syntax of the following with examples: 1. if statement 2. if else 3. if else ladder	*2*	*2*	*06*
B	Write a program to input sides if a triangle and check whether it is equilateral(All sides equal), isoceles(Two sides equal), scalene(No side equal)	*3*	*3*	*06*
C	Describe the use and syntax of Switch case. Write a program to input a single digit number and convert it into it's equivalent word. Eg: Input: 1 Output: One	*2,3*	*2,4*	*06*
Q4	**Attempt All**	***CO***	***B.T. Level***	***Marks***
A	Summarize the use and syntax of 1-Dimensional arrays. Write a program to declare and initialize and display contents of 1-D array	*3*	*2,3*	*06*
B	Write a short note on using Strings in C. Explain with the help of a program	*2,3,5*	*2,3*	*06*
Q5	**Attempt All**	***CO***	***B.T. Level***	***Marks***
A	Differentiate between union and structures	*4*	*3,4*	*06*
B	Explain the concept of pointers with a program.	*4,5*	*3*	*06*

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F.Y. B. Tech. NEP (Civil/Mech/CSE/IT) Examination

I S E –I (Class Test-I) Examination Oct 2024

MABSC 1001: Mathematics-I

Time: 1 Hour *Max. Marks:15* *Date: 25/10/2024* *Time:- 03.30 to 04.30*

"`Verify the Course Code and check whether you have got the Correct Question Paper"

N.B.

1. ***Attempt all questions.***
2. ***Figures to the right indicate full marks***
3. ***Assume suitable data if necessary and state it clearly***
4. ***Use of Non-Programmable Calculator and Data Sheet is allowed***

No.	Questions	COs	BT Levels	Marks
1.	Attempt any two			
	a) State De` Moivre`s theorem. Use it solve $x^6+x^5+x^4+x^3+x^2+x+1=0$	CO 1	K-1	**04**
	b) If $\tan(x+iy)=i$ where x *and* y are real. prove that x is indeterminate and y is infinite.			**04**
	c) If $\sin(a-ib)=x+iy$ then prove that $\frac{x^2}{\cosh^2 b}+\frac{y^2}{\sinh^2 b}=1$.	CO 2	K-2	**04**

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2	Attempt all questions a) Using echelon form , find the rank of A , where $A=\begin{bmatrix} 1 & 4 & 3 & 1 \\ -1 & -3 & 3 & 2 \\ 1 & 2 & 2 & 1 \\ 1 & -1 & 5 & -3 \end{bmatrix}$	CO 1	K-1	03
	b) Define rank of matrix ,using normal form ,find the rank of A $A=\begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 9 & 10 & 11 & 12 \end{bmatrix}$	CO 2	K-2	04
	OR c) Find non singular matrices P & Q such that PAQ is in normal form. Hence find rank of A, where $A=\begin{bmatrix} 1 & 2 & 0 & -1 \\ 3 & 4 & 1 & 2 \\ -2 & 3 & -2 & 5 \end{bmatrix}$.			04

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Q. N

Government College of Engineering, Chatrapati Sambhajinagar

(An Autonomous Institute of Government of Maharashtra)

F.Y. B. Tech. (All) (NEP) Examination

I S E –II Examination Dec 2024

MABSC 1001/ MABSC 1002: Mathematics-I

Time: 1 Hour *Max. Marks:15* *Date: 21/12/2024* *Time:- 03.00 to 04.00pm*

"Verify the Course Code and check whether you have got the Correct Question Paper"

N.B.

1. Attempt all questions. 2. Figures to the right indicate full marks 3. Assume suitable data if necessary and state it clearly 4. Use of Non-Programmable Calculator and Data Sheet is allowed

Q. No.	Questions	COs	BT Levels	Marks
Q.1.	Attempt any two			
	a) Define gamma function. Evaluate $\int_0^{\infty} x^9 \, e^{-2x^2} \, dx$	CO 1	K-1	04
	b) Write trigonometric form of Beta function. Evaluate $\int_0^{1} \frac{x^2}{\sqrt{1-x^5}} \, dx$	CO 2	K-2	04
	c) Evaluate $\int_0^{\infty} 7^{-4x^2} \, dx$			04

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Q.2.	Attempt all questions			
	a) State Leibnitz's theorem of successive differentiation If $y = e^{b\cos^{-1}x}$ then prove that $(1-x^2)y_{n+2} - (2n+1)x\,y_{n+1} - (n^2+b^2)y_n = 0$	CO 1	K-1	03
	b) Obtain the Maclaurin's expression of $\tan\left(x+\frac{\pi}{4}\right)$ and hence find the value of $\tan(47.5^0)$	CO 2	K-2	04
	OR			
	c) State Taylor's Theorem in ascending powers of $(x-a)$ & Express $(x+4)^4 - 3(x+4)^3 + 4(x+4)^2 + 5$ in powers of x using Taylor's theorem.			04

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Government College of Engineering, Chattrapati Sambhaji Nagar

FY.B.Tech ISE –I Examination Oct- 2024

CHBSC1001: ELECTROCHEMISTRY, BATTERY SCIENCE AND ENGINEERING MATERIALS

Time: 01 Hr. *Max. Marks: 15*

Q1	**Attempt any THREE of the following**	*CO*	*B.T. Level*	*Marks*
A.	I)Predict spontaneity of reaction - $Fe^{3+} + 2Cl^- \rightarrow Fe + Cl_2$, $E^0\ Fe^{3+}/Fe = -0.440$ volt ; $E^0\ Cl/Cl- = 1.36$ volt II) Calculate the emf of the cell, $Cu^{2+} + Fe \rightarrow Cu + Fe^{2+}$ $Cu \rightarrow Cu^{2+} + 2e-$; $E^0 = -0.34$ volts $Fe \rightarrow Fe^{2+} + 2e-$; $E^0 = 0.41$ volts	*CO4*	*K4*	*04*
B.	With figure explain construction and working of bomb calorimeter and derive equation for GCV ?	*CO2*	*K2*	*04*
C	Why gaseous fuels are superior than solid and liquid fuels explain with advantages? Write a note on solar cells?	*CO3*	*K3*	*04*
D	What is flame photometry? With figuerExplain in short and give applications?	*CO2*	*K3*	*04*
Q2	**SOLVE ONE OF THE FOLLOWING**	*CO*	*B.T.*	*Marks*
A.	Define GCV and NCV and explain why GCV is greater than NCV?	*CO2*	*K2*	*03*
B	State and explain characteristics of good fuels with explanation?	*CO1*	*K1*	*03*

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Government College of Engineering Aurangabad, Chattrapati Sambhaji Nagar

(An Autonomous Institute of Government of Maharashtra)

End Semester Examination January 2025

First Year B. Tech. **CHBSC1001: ELECTROCHEMISTRY, BATTERY SCIENCE AND ENGINEERING MATERIALS**

15 JAN 2025

Time: 2.30 min. ***Max. Marks: 60***

"Verify the Course Code and check whetheryou have got the Correct Question Paper"

N.B.

1. ***Attempt all questions.***
2. ***Figures to the right indicate full marks.***
3. ***Assume suitable data if necessary and state it clearly.***
4. ***Use of Non-Programmable Calculator and Data Sheet is allowed.***

Q.1	**Attempt any two**	***CO***	***B.T. Level***	***Marks***
A	Differentiate between SWCNT and MWCNT	***CO4***	***BT4***	***06***
B	With Reactions explain construction and working of Ni-Cd batteries Give the advantages and disadvantages of Ni-Cd batteries?	***CO1***	***BT1***	***06***
C	Define electrochemical series? Explain applications of electrochemical series in detail?	***CO3***	***K3***	***06***
Q.2	**Attempt any two**	***CO***	***B.T. Level***	***Marks***
A	What is ultra sound assisted organic reactions? Explain with example?	***CO2***	***BT2***	***06***
B	Wt. of coal sample = 0.92 g; wt. of water = 550 g; water equivalent of calorimeter = 2,200 g; temperature rise (t2-t1) = 2.42 C; acid correction = 50.0 cal; fuse wire correction = 10.0 cal; latent heat of steam = 587 cal/g; percentage of H = 6%. Calculate GCV and NCV? Why NCV is lower than GCV?	***CO4***	***BT4***	***06***
C	What are conducting polymer? Give classification of conducting polymer? Explain mechanism of conduction in Polyacetyelene?	***CO2***	***BT2***	***06***
Q.3	**Attempt any two**	***CO***	***B.T. Level***	***Marks***
A	List twelve principals of green chemistry? Explain any two in detail?	***CO2***	***BT2***	***06***
B	Explanatory notes on i Fullerenes ii.EV battery	***CO2***	***BT2***	***06***
C	Explain classification of LCP in details? Give their applications?	***CO2***	***BT2***	***06***
Q.4	**Attempt any two**	***CO***	***B.T. Level***	***Marks***
A	Explain following (any two) i Capacity ii Voltage (Cell Potential) iii Cycle Life	***CO1***	***BT1***	***06***
B	With figure Discuss the principles and working of Hydrogen Oxygen fuel cell with cell reactions?	***CO2***	***BT2***	***06***

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C	Describe the terms i) Equivalent conductance ii) specific conductance iii) molar conductance Predict the following cell reaction is feasible or not. Zn/Zn^{2}+ (1M)//Cu^{2}+/Cu; Given that E^{o} Cu^{2+}/Cu=0.34V and E^{o} Zn^{2}+/Zn= -0.76V.	*CO3*	*BT3*	*06*
Q.5	**Attempt any two**	*CO*	*B.T. Level*	*Marks*
A	Describe construction and working OF Li-ion batteries with chemical reactions during charging and discharging?	*CO2*	*BT2*	*06*
B	What are biodegradable polymers? Give synthesis, properties and applications PLA?	*CO3*	*BT3*	*06*
C	$CH_4(g) + H_2O(g) ==> 3H_2(g) + CO(g)$ $Fe_2O_3(s) + 3CO(g) ===> 2Fe(l) + 3CO_2(g)$ Calculate atom economy of this reaction?	*CO5*	*BT5*	*06*

Note: CO: Course Outcome; B.T. Level: Bloom's Taxonomy Level

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Government College of Engineering Aurangabad,
Chhatrapati Sambhajinagar
(An Autonomous Institute of Government of Maharashtra)
***F.Y. B.Tech (C/M/CSE/IT)* Examination (NEP)**
End Semester Examination January 2025
MABSC 1001: Mathematics-I

Time: 2Hrs and 30Min. Date of Examination: 13 JAN 2025 *Max. Marks: 60*

"Verify the course code and check whether you have got the correct question paper"

N.B:-

1. *Assume suitable data if necessary and state it clearly*
2. *Use of non-programmable calculator is allowed*
3. *Attempt all questions in sequence*

Q. No.	Questions	Course Outcomes (Cos)	Bloom's Taxonomy Levels (BTLs)	Marks
Q1.	Solve any three a) Show that $(1+\cos\theta+i\sin\theta)^n+(1+\cos\theta-i\sin\theta)^n=\left(2\cos\frac{\theta}{2}\right)^n\left(2\cos\frac{n\theta}{2}\right)$ b) Solve the equation $x^{10}+11x^5+10=0$. c) Prove that $\operatorname{sech}^{-1}(\sin\theta)=\log\cot\left(\frac{\theta}{2}\right)$. d) Separate real and imaginary parts of $\sin^{-1}(\operatorname{cosec}\theta)$.	CO3, CO5	K2	12
Q2.	Solve any two a) Find non singular matrices P *and* Q such that PAQ is in normal form. Also find its rank; where $A=\begin{bmatrix}1&2&3&4\\2&1&4&3\\3&0&5&-10\end{bmatrix}$. b) Find the value of p for which the following matrix A will be of (i) rank one (ii) rank two (iii) rank three $A=\begin{bmatrix}3&p&p\\p&3&p\\p&p&3\end{bmatrix}$.	CO2, CO3, CO5	K2	12

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	c) Solve the following system of equations : $3x-y-z=0$ $x+y+2z=0$ $5x+y+3z=0$			
Q3.	Solve any three a) Find n^{th} derivative of $y=\cos^5 x\sin^3 x$. b) If $y=a\cos(\log x)+b\sin(\log x)$, prove that $x^2y_{n+2}+(2n+1)xy_{n+1}+(n^2+1)y_n=0$. c) State Taylor's series and expand $\tan^{-1}x$ in powers of $(x-1)$. d) Prove that $e^{a\sin^{-1}x}=1+ax+\frac{a^2x^2}{2!}+\frac{a(a^2+1)}{3!}x^3+\cdots\cdots$	CO2, CO4	K2	12
Q4.	Solve any three a) Evaluate $\int_0^\infty \frac{x^7}{7^x}dx$. b) Evaluate $\int_0^1 x^4\log\left(\frac{1}{x}\right)^3 dx$. c) Evaluate $\int_0^\infty \frac{e^{-\beta x}\sin\alpha x}{x}dx$ by using rule of differentiation under integral sign and hence deduce that $\int_0^\infty \frac{\sin\alpha x}{x}dx=\begin{cases}-\frac{\pi}{2} & if \quad \alpha<0\\ 0 & if \quad \alpha=0\\ \frac{\pi}{2} & if \quad \alpha>0\end{cases}$ d) Show that $erf(\infty)=1$ *and* $erf(0)=0$.	CO2, CO4	K2	12
Q5.	Solve any two a) Trace the strophoid with full justification $y^2(a+x)=x^2(b-x)$. b) Trace the Folium of Descartes with full justification $x^3+y^3=3axy$. c) Trace the curve with full justification $r=a(1-\cos\theta)$..	CO5	K3	12

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Government College of Engineering Aurangabad, Chhatrapati Sambhaji Nagar

(An Autonomous Institute of Government of Maharashtra)

End Semester Examination November December 2024

First year B. Tech. (Electronics & Telecommunication/ Computer Science Engg) – Semester – I

CSESC 1001: Programming for Problem Solving (PPS)

Time: 3Hours ***Max. Marks: 60***

"Verify the Course Code and check whether you have got the Correct Question Paper"

N.B.

1. ***Attempt all questions.***
2. ***Figures to the right indicate full marks.*** 10 JAN 2025
3. ***Assume suitable data if necessary and state it clearly.***
4. ***Use of Non-Programmable Calculator and Data Sheet is allowed.***

Q.1	**Attempt any six from following**	***CO***	***BTL***	***Marks***
A	Differentiate between compiler and interpreter	***CO1***	***U***	***02***
B	Give meaning of syntax error and logical error? Give one example of each.	***CO1***	***R***	***02***
C	Define operator? Give meaning of unary, binary and tertiary operator. Give one example of each.	***CO2***	***U***	***02***
D	State the difference between break and continue statement. How these are executed? Write one example of it	***CO3***	***U***	***02***

E	Correct the error in following code if any?
	`void main()`
	`{ inti=1;`
	`do`
	`print("%d",i);`
	`i=i+1;`
	`while(i<=10)`
	`}`
	
	`int main(intarg. char argv)`
	`{ inti;`
	`for(i=1,i<argc,i++)`
	`printf("%d ",argc);`
	`}`
		CO3	***A***	***02***

F	Assuming x=5, y=0 and z=1 initially, what will be values of x,y,z after executing the following code segment
	`if ( x )`
	`if( y )`
	`z=10;`
	`else`
	`z=0;`
	
	`if ( x && z)`
	`x=10;`
	`else`
	`y=10;`
		CO4	***A***	***02***

G	(i) char name[5]; name="GECA"; (ii) char *name; name="GECA" which statement is correct? Justify your answer.	***CO5***	***U***	***02***
H	Write prototype of user defined function having name "power"which has two arguments one is "base" of float type and other is "index" of integer type. Function returns the value of double type	***CO3***	***U***	***02***
Q.2	**Attempt any four**	***CO***		***Marks***
A	Write a program to determine given character is alphabet, digit or special character	***CO2***	***A***	***03***
B	Give syntax and purpose of (i) strcmp() (ii) strcpy (iii) strcat function. Give one example of each	***CO4***	***U***	***03***

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C	What is the output of the following program
	`#include <stdio.h>`
	`int x = 5;`
	`void print();`
	`int main()`
	`{ int x = 3; printf("\n%d", x);`
	`{ staticint x; printf("\n%d",x); }`
	`print();`
	`printf("\n%d", x);`
	`}`
	`void print()`
	`{ x++; printf("\n%d",x); }`
		CO3	A	03

D	If array is declared as float number[25]; write (1) type of array (ii) name of array (iii) number of elements of array (iv) size of the array v) dimension of the array vi) assign value 1 to all elements of array	CO4	U	03
E	Write code segment to find larger of two numbers x and y using (i) if only (ii)if else (iii) conditional operator	CO3	A	03
Q.3	**Attempt any three**	CO		Marks
A	State difference between arrays and structures (any four points)	CO4 CO5	U	04
B	What is nested loops? Give its syntax? How it operates? State its applications?	CO3	U	04
C	**Write user defined function int leap (int) which checks given year is leap or not and return value accordingly to calling function. Calling function shall print the year as 'leap year' or 'not leap year'**	CO3	A	04
D	Define (i) algorithm (ii) flowchart. Write algorithm and draw flowchart for a program to determine given number is odd or even.	CO1	U	04
Q.4	**Attempt any two from following**	CO		Marks
A	**Write a program to accept a 3x3 matrix from keyboard and print the given matrix and transpose of it.**	CO4	A	06
B	**What is recursion? What are its advantages and disadvantages over iteration? Explain recursive function with suitable example**	CO3	U	06
C	**What is sorting? List different types of sorting techniques. Illustrate any one type of sorting technique with relevant 'C' program**	CO4	A	06
Q.5	**Attempt any two from following**	CO		Marks
A	**With suitable example explain the concept of call by value and call by reference. Write its importance.**	CO3 CO5	U	06
B	Write a program in 'C' to accept and print information of five students such as (i) Name of student (ii) Gender as 'M' or 'F' (iii) Enrolment Number iv) Percentage of marks	CO5	A	06
C	Write a program to accept a string. Find the length of a string. Print the string in reverse order using pointer.	CO5	A	06

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Government College of Engineering, Aurangabad

(An Autonomous Institute of Government of Maharashtra)

F.Y. B.Tech (C/M/CSE/IT) Examination (NEP)

End Semester Examination December 2023

MABSC 1001: Mathematics-I

Time: 2Hrs and 30Min. Date of Examination: 18 DEC 2023 *Max. Marks: 60*

"Verify the course code and check whether you have got the correct question paper"

N.B:-

1. *Assume suitable data if necessary and state it clearly*
2. *Use of non-programmable calculator is allowed*
3. *Attempt all questions in sequence*

Q. No.	Questions	Course Outcomes (Cos)	Bloom's Taxonomy Levels (BTLs)	Marks
Q1.	Solve any three a) Expand in polar form $\frac{(1+i)^8(\sqrt{3}-i)^4}{(1-i)^4(\sqrt{3}+i)^8}$. b) Find all the roots of $\left(\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)^{\frac{3}{4}}$. c) Prove that $\sin^5\theta = \frac{1}{16}(\sin 5\theta - 5\sin 3\theta + 10\sin\theta)$. d) Separate real and imaginary parts of $\sin^{-1}(\operatorname{cosec}\theta)$.	CO3, CO5	K2	12
Q2.	Solve any two a) Find nonsingular matrices P and Q such that PAQ is in normal form. Also find its rank; where $A=\begin{bmatrix}1 & 2 & -3 & -2\\ 2 & -3 & 1 & 2\\ -1 & 1 & 3 & 2\end{bmatrix}$. b) Solve the following system of equations : $x+y+z=6$ $x+2y+3z=14$ $2x+4y+7z=30$ c) Find eigen values and corresponding eigen vectors for the matrix $A=\begin{bmatrix}3 & -1 & 1\\ -1 & 5 & -1\\ 1 & -1 & 3\end{bmatrix}$.	CO2, CO3, CO5	K2	12

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Q3.	Solve any three a) Find n^{th} derivative of $y=\cos^3 x \sin^5 x$. b) If $y=\left(x^2-1\right)^n$, prove that $\left(x^2-1\right)y_{n+2}+2xy_{n+1}-n(n+1)y_n=0$. c) Expand $\log(\cos x)$ about $\frac{\pi}{3}$. d) Prove that $\sinh^{-1}x = x-\frac{x^3}{6}+\frac{3x^5}{40}-\ldots\ldots$	CO2, CO4	K2	12
Q4.	Solve any three a) Evaluate $\int_0^\infty \frac{x^5}{5^x}dx$. b) Evaluate $\int_0^1 \frac{x^2}{\sqrt{1-x^4}}dx \cdot \int_0^1 \frac{dx}{\sqrt{1-x^4}}$. c) Evaluate $\int_0^\infty \frac{e^{-x}}{x}\left[1-e^{-ax}\right]dx, \quad (a>-1)$. d) Show that $erf(\infty)=1$.	CO2, CO4	K2	12
Q5.	Solve any two a) Trace the curve with full justification $xy^2=4a^2(a-x)$. b) Trace the curve with full justification $y^2(a+x)=x^2(b-x)$. c) Trace the curve with full justification $r^2=a^2\cos 2\theta$.	CO5	K3	12

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Government College of Engineering Aurangabad, Chhatrapati Sambhaji Nagar

(An Autonomous Institute of Government of Maharashtra)

End Semester Examination December-2024

First Year B. Tech CIVIL & CSE. – NEP – Semester – I

CEESC1001: Basic of Civil Engineering

Time: One and Half Hour 8 JAN 2025 *Max. Marks: 30*

"Verify the Course Code and check whether you have got the Correct Question Paper"

N.B.

1. *Attempt all questions.*
2. *Figures to the right indicate full marks.*
3. *Assume suitable data if necessary and state it clearly.*
4. *Use of Non-Programmable Calculator and Data Sheet is allowed.*

Q.1	**Attempt any two**	*CO*	*B.T. Level*	*Marks*
A	What is Foundation? Explain its function & any one types of foundation in details with neat sketch.	*CO1*	*K2*	*05*
B	Distinguish between English bond & Flemish bond.	*CO1*	*K1*	*05*
C	Explain in details various types of structure with neat sketch.	*CO1*	*K2*	*05*
Q.2	**Attempt any two**	*CO*	*B.T. Level*	*Marks*
A	Explain Principle of Surveying & Types of surveying.	*CO2*	*K2*	*05*
B	Write details note on any Two Modern survey equipment's.	*CO2*	*K1*	*05*
C	Convert following WCBs to QBs/RBs a) WCB of AB = $45^0 30'$ b) WCB of BC= $125^0 45'$ c) WCB of CD= $222^0 15'$ d) WCB of DE= $320^0 30'$	*CO2*	*K3*	*05*
Q.3	**Attempt any two**	*CO*	*B.T. Level*	*Marks*
A	What is Cement? Explain Properties & Field tests of cement.	*CO3*	*K1*	*05*
B	Define Superelevation. Explain its Objective & design consideration.	*CO3*	*K2*	*05*
C	Explain classification of Dams in details.	*CO4*	*K2*	*05*

Note: CO: Course Outcome; B.T. Level: Bloom's Taxonomy Level

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